

1. PREFACE

Thank you for purchasing our temperature sensor.

This User's Manual gives instructions on handling, wiring, and maintenance of temperature sensors.

In order for the temperature sensor to be fully functional and to operate in an efficient manner, the manual must be carefully read.

■ Notes on the User's Manual

- In no case does this manual guarantee the merchantability of the transmitter or its adaptability to a specific client need.
- Changes to specifications, structure, and components used may not lead to the revision of this manual unless such changes affect the function and performance of the products.

■ Warranty

- The warranty period of the instrument is written on the estimate sheet that is included with your purchase. Any trouble arising during the warranty period shall be repaired free of charge.
- Whether or not the defective instrument is repaired free of charge depends on the result of our inspection.
- **The following conditions shall not be eligible for charge-exempt repair.**
 - Problems caused by improper or insufficient maintenance on the part of the customer.
 - Trouble or damage caused by mishandling, misuse, or storage that exceeds the design or specification requirements.
 - Trouble or damage was caused by modification or repair that was handled by a party or parties other than our consigned agent.
 - Trouble or damage was caused by fire, earthquake, wind or flood damage, lightning strikes or other acts of God that are not directly a result of problems with this instrument.

■ Checking items after unpacking

When the temperature sensor is delivered, check the appearance for damage, and also verify that the configuration indicated in the "Model and Suffix Code" is in compliance with the specifications written on the order sheet.

■ Handling cautions for intrinsically safe type

TIIS intrinsically safe type (i3aG5)

Type	TC	TD	RB
Certification number	T28123	T28124	T28125
Combined barrier	BARD200	Two or less BARD200 Lext ≤ 2.2 mH Cext ≤ 1 μF	BARD300
Internal inductance Lint	≤ 20 μH		
Internal capacitance Cint	≤ 2 nF		
Ambient temperature Ta	-10°C ≤ Ta ≤ 55°C (-14°F ≤ Ta ≤ 131°F)		



CAUTION

Do not change the circuit or parts. The instrument modification will void the explosion proof certification.

2. Notes on Safety

It is absolutely imperative that users of this equipment read and observe the safety instructions mentioned in this manual in order to ensure the protection and safety of operators.

2.1 Installation

- Lead cables linked with the temperature sensor, such as a compensating lead cable or extension lead cable, must be connected to terminals of receiving instruments. An incorrect connection may cause a fire or an explosion.
- When the temperature sensor is dropped, people may be injured since some temperature sensors are heavy. Pay attention to the handling of temperature sensors, especially resistance thermometers with ceramic insulators or protections, which are very delicate.
- Temperature sensors of flameproof type are approved when that they are used under 60°C or below, and that of temperature for Intrinsically safe type is 55°C or below. Confirm the ambient temperature is in accordance with those conditions.
- After wiring, confirm if O-ring packing is furnished and then close the cover surely in order to prevent permeation of dust or water. Especially for the explosion proof type, fasten the cover firmly using a hexagonal 17 mm wrench. Otherwise, it may be declined the capability of the explosion proof.
- Screws or flanges are used for installation of temperature sensors. In case of screwing, apply sealing tape or sealing agent for tapered thread and apply O-ring gaskets for parallel thread. Then fasten them firmly. In case of flange connection, insert specified gaskets and screw up all bolts equally.

- The sheath can be bent providing that the bending radius is larger than the twice of the sheath diameter. However, the sheath will be damaged if it changes the form again. When you bend the sheath in the field, we recommend that the bending radius keep larger than five times of the sheath diameter. The metal sheathed resistance thermometers have resistance element in the top of the sheath. Therefore, do not bend the portion of the sheath with 100 mm length from the top.
- If the connection part of temperature sensors and lead cables are exposed to high temperature, the insulation resistance may be declined or the measuring error may be increased. Therefore, keep the temperature of the part at 80°C or less except products for high temperature application.
- If the polarity is incorrect, it is impossible to measure the correct temperature. Pay attention the color of compensating cables because those colors change with sensor type.

2.2 Maintenance and Inspection

- Flameproof sensors shall not be maintenance-serviced with its power turned on. However, in cases where maintenance servicing is to be conducted with the power turned on, with the terminal head cover removed, always use a gas detector to check that there is no explosive gas in that location. Do not attempt to change specifications about flameproof sensors.
- Since there are many cases of high temperature in the plant where temperature sensors are installed, it is very dangerous to maintain or inspect temperature sensors during operation or just after shutdown.
- If the temperature sensors are used as footing, they might be damaged.

2.3 Storage

- Store temperature sensors in dry and clean place. When temperature sensors are kept in the place of high humidity, the insulation resistance of them may decline.

3. Wiring

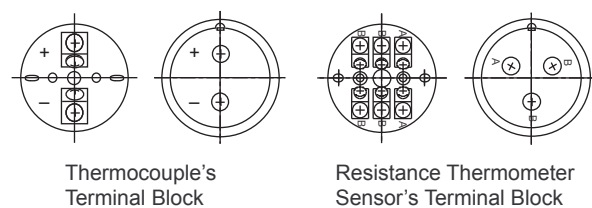
Note that lead cables to connect temperature sensor for thermocouple and for resistance are different.

(1) Thermocouples

For the extension wires, use the standard compensation lead wires recommended for each thermocouple type. Incorrect use of compensation wire type causes a large error. The color of wire covering changes with type of compensation lead wires and standard such as JIS, ASTM, IEC, etc. Also pay attention to the polarity of compensation wires when connect them.

(2) Resistance thermometer sensor

Since 3-wire method is adopted in resistance thermometer measurement usually, each sensor requires 3 wires. The terminal symbols are described as A, A, and B.



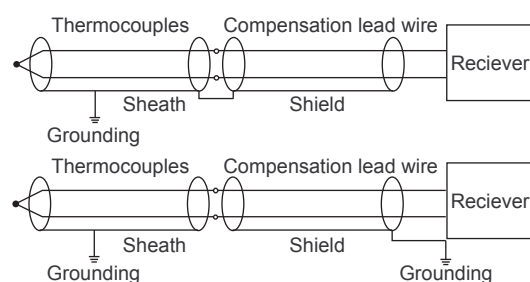
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Figure 1. Terminal Blocks

4. Grounding

In case the temperature sensor is connected with a receiver, a shield cable may be grounded in order to avoid the influence of the noise. Generally, one-point grounding is recommended.

Please refer to Figure 2 about the grounding points.



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Figure 2. Grounding of Shield

<http://www.yokogawa.com>